MCA Semester – IV Project

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WealthAI: AI-Powered Personal Finance Assistant with Machine Learning Classification, Spending Outlier Analytics, and Stochastic Projections

Research Project submitted to Jain Online (Deemed-to-be University)

In partial fulfillment of the requirements for the award of

Master of Computer Applications

Submitted by

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DECLARATION

I, Deviya HarshitKumar KalpeshKumar, hereby declare that the Research Project Report titled "WealthAI: AI-Powered Personal Finance Assistant with Machine Learning Classification, Spending Outlier Analytics, and Stochastic Projections" has been prepared by me under the guidance of Prof. Prabha S.

I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of the degree of Master of Computer Applications by Jain University, Bengaluru.

I have undergone a project for a period of Eight Weeks.

I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place: Bengaluru

Date: 13-08-2026

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CERTIFICATE

This is to certify that the Project report submitted by Mr. Deviya HarshitKumar KalpeshKumar bearing USN 241VMTR01558 on the title "WealthAI: AI-Powered Personal Finance Assistant with Machine Learning Classification, Spending Outlier Analytics, and Stochastic Projections" is a record of project work done by him during the academic year 2025-26 under my guidance and supervision in partial fulfilment of Master of Computer Applications.

Place: Bengaluru _____

Date: 13-08-2026 Prof. Prabha S

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to Jain Online (Deemed-to-be University) for providing the infrastructure and the opportunity to undertake this research project.

I express my deepest gratitude to my faculty guide, Prof. Prabha S, for her continuous support, technical insight, constructive critiques and patient guidance throughout the design and implementation phases of WealthAI.

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Executive Summary

Modern personal financial management is heavily fragmented. Individuals hold assets and maintain transaction accounts across multiple banking institutions, brokerage platforms and digital wallet applications. This division creates a significant barrier to maintaining an accurate, consolidated view of total net worth. Additionally, manual transaction categorisation is inefficient and prone to human error, unusual spending charges frequently remain undetected until statement periods close, and conventional long-term goal calculators rely on static returns that fail to account for real-world market volatility.

To solve these issues, WealthAI was developed as a full-stack, three-tier web application combining a MERN architecture (MongoDB Atlas, Express.js, React.js, Node.js) with a dedicated Python Flask microservice for machine learning and stochastic modelling. WealthAI automatically categorises imported bank-statement CSV records using Natural Language Processing (NLP) paired with a Naive Bayes classifier. It flags irregular spending through an Isolation Forest model (backed by statistical Z-score fallback logic), generates short-term expense forecasts via linear regression, projects investment goals using Monte Carlo simulations across 1,000 iterations, tracks asset portfolios via live market data from the yfinance API, and offers conversational assistance through a built-in financial chatbot.

The system was evaluated using both primary user data (CSV uploads, manual ledger entries, budget targets) and secondary market data. The platform is deployed across cloud infrastructure: Vercel hosts the React frontend, Render hosts the Express API backend and the Flask ML microservice, and MongoDB Atlas manages persistent storage. This report presents the requirement analysis, system architecture, mathematical formulations, database schemas, API routes, operational workflows, empirical findings and future development paths for WealthAI.

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CHAPTER 1

INTRODUCTION, SCOPE AND BACKGROUND

1. INTRODUCTION, SCOPE AND BACKGROUND

1.1 Overview of Project Case / Business case

Personal finance management is an increasingly complex responsibility for modern households. The financial ecosystem offers a wide variety of services across traditional retail banks, equity brokerages, mutual fund platforms and digital payment gateways. While this expansion gives consumers options, it also fragments their financial data. Users must log into multiple separate portals to evaluate their balances, track debts, verify recurring subscriptions and track overall wealth growth.

Managing personal finances without unified tools leads to blind spots. Individuals frequently fail to track discretionary spending trends across multiple accounts, leading to missed savings targets. Manual tracking methods, such as spreadsheets or pen-and-paper ledgers, are labour-intensive, error-prone and lack real-time predictive capabilities. Furthermore, conventional financial advice platforms often use static interest formulas that assume fixed asset growth over decades, ignoring market volatility.

The business case for WealthAI focuses on closing the gap between raw financial transaction tracking and actionable predictive analytics. By consolidating multi-account data into a single interactive dashboard and applying machine learning to automated expense categorisation, outlier detection and stochastic projections, WealthAI gives users clear financial insights. Designed within the data analytics domain for the Master of Computer Applications curriculum at Jain Online (Deemed-to-be University), this project demonstrates how modern web software design can integrate with data science algorithms.

1.2 Problem definition

The core challenges in personal finance management stem from three main issues:

- Information Fragmentation across Financial Entities: Consumers maintain separate accounts across multiple retail banks, investment brokers, credit card issuers and digital wallets. Without a single aggregating platform, computing total net worth requires manual labour.
- High Latency and Errors in Manual Categorisation: Expense classification is essential for budgeting. Manually assigning categories to dozens or hundreds of raw monthly transaction descriptions is slow and inconsistent. As a result, budget tracking often breaks down due to user fatigue.

- Delayed Anomaly Recognition and Static Long-Term Forecasting: Unusual transactions - such as accidental double charges, rate hikes on subscriptions or fraudulent activity - often go unnoticed until month-end reviews. Additionally, standard goal calculators use static interest rates A = P(1 + r)^t that do not account for real-world market fluctuation.

WealthAI addresses these challenges directly by delivering an automated software platform capable of extracting, processing, categorising, analysing and forecasting personal finance data.

1.3 Project Scope

1.3.1 Aim of the Project

The primary objective of WealthAI is to design, implement and deploy an automated, full-stack, AI-driven personal finance application. The system consolidates user records, categorises incoming transactions via machine learning, flags irregular spending patterns, evaluates long-term goals using stochastic Monte Carlo simulations, tracks live asset prices and provides real-time financial assistance via a web interface.

1.3.2 Functional Scope

The functional scope of the project encompasses seven core modules:

- Automated Transaction Import and ML Categorisation: Ingests user-uploaded bank-statement CSV files, parses raw text descriptions and applies a Natural Language Processing (NLP) classifier to assign financial categories automatically.
- Budgeting, Savings Goals and Subscription Tracking: Monitors user-defined spending limits per category, tracks target savings milestones and raises alerts for recurring subscription renewals.
- Portfolio Management with Live Market Quotes: Tracks equities, ETFs and other assets, integrating external market feeds to update total net worth in real time.
- Spending Anomaly Detection: Evaluates incoming transaction amounts using unsupervised isolation models and statistical threshold checks to highlight unexpected spending outliers.
- Short-Term Expense Forecasting: Uses historic spending trend analysis to project expected outlays for upcoming billing cycles.

- Stochastic Goal and Portfolio Projections: Runs Monte Carlo simulations (1,000 iterations per execution) based on historical asset returns and standard deviations to model realistic probability distributions for future wealth.
- Interactive Financial Chatbot: Features an integrated text chatbot to answer user enquiries regarding budget balances, savings strategies and portfolio status.

1.3.3 Core Machine Learning Specifications

Table 1.1: Machine Learning Algorithms and Mathematical Hyperparameters

System Function	Machine Learning / Statistical Model	Specific Parameters and Pipeline Configuration	Operational Fallback Rule
Transaction Categorisation	Multinomial Naive Bayes (MultinomialNB)	TfidfVectorizer(lowercase=True, stop_words='english', ngram_range=(1,2)) coupled to MultinomialNB(alpha=0.1)	Rule-based exact keyword dictionary matching.
Anomaly Detection	Isolation Forest (IsolationForest)	IsolationForest(n_estimators=100, contamination=0.05, random_state=42)	Z-score statistical threshold check (Z > 2.5) when transaction count < 15.
Expense Forecasting	Ordinary Least Squares Linear Regression	LinearRegression() trained on monthly historic aggregate expenditures	Rolling historic moving average when data span < 3 months.
Portfolio / Goal Projection	Monte Carlo Simulation	N = 1000 iterations driven by Gaussian asset distribution (mean mu, standard deviation sigma)	Deterministic compound interest modelling.
Market Data Retrieval	Python Financial Gateway	yfinance API fetching real-time ticker prices and market search results	Cached historic close quotes.

CHAPTER 2
REVIEW OF LITERATURE

2. REVIEW OF LITERATURE

2.1 Literature Review

Personal financial software has evolved from simple digital ledgers to complex, predictive systems. Early software relied entirely on manual entry and rule-based calculations. While effective for basic record-keeping, these systems placed a heavy operational burden on users.

Recent advances in natural language processing (NLP) and supervised classification have streamlined financial data ingestion. Text vectorisation methods, such as Term Frequency-Inverse Document Frequency (TF-IDF), extract key features from raw, unstructured transaction strings. Paired with Naive Bayes classifiers, these pipelines map vague merchant descriptions (for example, "AMZN MKTPLACE PMTS") to distinct spending categories (for example, "Shopping") with low computational overhead.

In anomaly detection, traditional personal finance tools relied on fixed thresholds, such as flagging any transaction over a set amount. However, fixed thresholds produce high false-positive rates because they fail to adjust to individual spending patterns. Machine learning techniques such as Isolation Forests isolate anomalies by randomly selecting features and splitting data values. Because anomalies require fewer splits to isolate than normal data points, they appear significantly higher up the tree structure. For sparse datasets, statistical metric bounds such as the Z-score offer a reliable fallback for catching statistical outliers without overfitting.

Finally, deterministic investment planning models assume fixed annual interest rates, ignoring market volatility. Stochastic modelling, specifically Monte Carlo simulation, addresses this by running thousands of potential market trajectories sampled from historic mean returns (mu) and standard deviations (sigma). This produces a realistic range of outcomes - including 10th, 50th and 90th percentile projections - giving users a clearer view of financial risk.

2.2 Feasibility Analysis

2.2.1 Technical Feasibility

The proposed system uses a three-tier architecture: a React.js single-page application for the user interface, a Node.js/Express.js API for backend business logic, and a Python Flask microservice for machine learning and statistical modelling. Communication relies on standard HTTP REST protocols. Data persistence is handled by MongoDB Atlas via the Mongoose Object Data Modelling (ODM) framework. Modern cloud infrastructure - including Vercel for

frontend hosting and Render for backend service deployment - provides the computational resources required for smooth operation. Technical feasibility is therefore fully established.

2.2.2 Economic Feasibility

The platform was built primarily using open-source libraries, frameworks and free-tier cloud services. Production deployment costs were minimised by leveraging Vercel (frontend tier), Render (backend and ML service tiers) and MongoDB Atlas (database cluster). Because the operational infrastructure uses zero-cost tier allocations during development and testing, the project is highly economically feasible.

2.2.3 Operational Feasibility

WealthAI features an intuitive user interface built with Material UI and Recharts. The system simplifies user workflows by automating manual steps: bank statements are uploaded via standard CSV files, transaction categorisation is automated, anomalous charges are highlighted automatically and portfolio performance is visualised through clean charts. Because the platform reduces manual data entry and requires no technical expertise to operate, operational feasibility is high.

2.2.4 Ethical Feasibility

Financial data requires strict security and privacy protections. WealthAI handles credential security by hashing user passwords with bcryptjs (salt factor 10) prior to database insertion. State authentication relies on JSON Web Tokens (JWT) transmitted via secure headers. To protect the system architecture, the Python Flask ML service is reached only through an internal service endpoint and is never exposed to public browser traffic. All client traffic routes exclusively through the Node.js API layer, ensuring secure multi-tenant data isolation.

CHAPTER 3
PROJECT PLANNING AND METHODOLOGY

3. PROJECT PLANNING AND METHODOLOGY

3.1 Project Planning

Development followed an eight-week timeline divided into distinct technical phases. Table 3.1 details the weekly sprint breakdown.

Table 3.1: Eight-Week Agile Sprint Schedule

Week	Technical Sprint Focus	Deliverables and Milestone Activities
Week 1	Requirement Gathering and Architecture Setup	Finalised system specifications, set up Git repositories, configured environment configurations.
Week 2	Database and Schema Engineering	Created Mongoose schemas for User, Transaction, Budget, Goal, Holding and PortfolioTransaction models.
Week 3	Authentication and Core API Layer	Implemented JWT authorisation middleware, bcrypt password hashing and user auth endpoints.
Week 4	Ledger and Budgeting Engine	Built CSV file upload handling, manual transaction entry routes and category budget calculations.
Week 5	ML Microservice Development	Trained the TF-IDF Naive Bayes categoriser, built the Isolation Forest anomaly detector and created Flask REST endpoints.
Week 6	Portfolio and Market Data Pipeline	Integrated yfinance API endpoints, holding calculations and Monte Carlo stochastic projection algorithms.
Week 7	Chatbot and Analytics Dashboard	Developed conversational routing logic, integrated Recharts UI visualisations and optimised UI flows.
Week 8	Integration Testing and Cloud Deployment	Executed end-to-end integration tests and deployed services to Vercel, Render and MongoDB Atlas.

Table 3.2: Risk Identification, Impact Assessment and Mitigation Strategies

Risk Identification	Probability / Impact	Technical Mitigation Strategy
Cold-start delays on serverless deployment platforms (Render free tier).	Medium / Low	Implemented lightweight ping health checks (GET /health) during startup to wake services before user interaction.
ML categoriser unreliability on sparse or unseen user datasets.	Medium / Medium	Built a rule-based dictionary fallback mechanism to assign standard category fallbacks for low-confidence scores.
Unstable external market data responses from public APIs.	Low / High	Added exception-handling wrappers around yfinance

		endpoints to return cached historic quotes if live fetches time out.

3.2 Methodology

An Agile software development methodology was selected for WealthAI. Two alternatives were considered. The Waterfall model was rejected because its sequential, single-pass structure does not tolerate the requirement churn that arises when machine learning behaviour is validated against real transaction data. The Spiral model was rejected as disproportionate to an eight-week, single-developer academic project because of its heavy risk analysis overhead per cycle.

Agile's iterative approach allowed continuous integration between the React frontend, the Node.js API backend and the Python ML service. Weekly sprints ensured that core API endpoints were built and tested before dependent frontend modules were connected, reducing integration issues across microservice boundaries.

CHAPTER 4
DATA ANALYSIS, DESIGN AND IMPLEMENTATION

4. DATA ANALYSIS, DESIGN AND IMPLEMENTATION

4.1 Requirement Analysis

4.1.1 Data Collection

The system uses both primary and secondary data sources:

- Primary Data: Financial transactions entered manually by the user or uploaded via bank-statement CSV files, along with user-defined target budgets, long-term financial goals and portfolio holdings.
- Secondary Data: Real-time market prices, historical asset performance and ticker search results fetched directly from the yfinance API.

4.1.2 Data Analysis and tools of data analysis

Data analysis in WealthAI is performed by the Python Flask microservice using scikit-learn, pandas and NumPy. Raw transaction descriptions are vectorised with TF-IDF and classified with Multinomial Naive Bayes; transaction amounts are screened for outliers with an Isolation Forest; monthly aggregates are regressed linearly to forecast the next cycle's expenditure; and goal projections are produced by Monte Carlo sampling. The exact parameters of each pipeline are given in Table 1.1. Results are returned to the Express API as JSON and rendered in the React client as Recharts tables and charts.

Functional Requirements

- FR-1: Secure authentication via JWT upon successful password validation.
- FR-2: Automated CSV parsing and classification of transaction records.
- FR-3: Real-time spending anomaly detection using Isolation Forest models and Z-score fallbacks.
- FR-4: Stochastic goal and net worth projections using 1,000-iteration Monte Carlo simulations.
- FR-5: Conversational financial queries answered via an integrated chatbot microservice.

Non-Functional Requirements

- NFR-1 (Security): Password storage protected using bcryptjs salt hashing; all protected routes guarded by JWT middleware.
- NFR-2 (Performance): Client dashboard visual updates render in under two seconds.
- NFR-3 (Database): MongoDB Atlas collections indexed on the user reference so that every query is scoped to a single authenticated account.
- NFR-4 (Maintainability): Strict separation of concerns across the three tiers, with the ML service replaceable without any frontend change.

4.2 Design and Architecture

WealthAI uses a decoupled, three-tier architecture to maintain a strict separation of concerns, as shown in Figure 4.1. The presentation tier never contacts the machine learning tier directly; all traffic is proxied through the authenticated Express application tier.

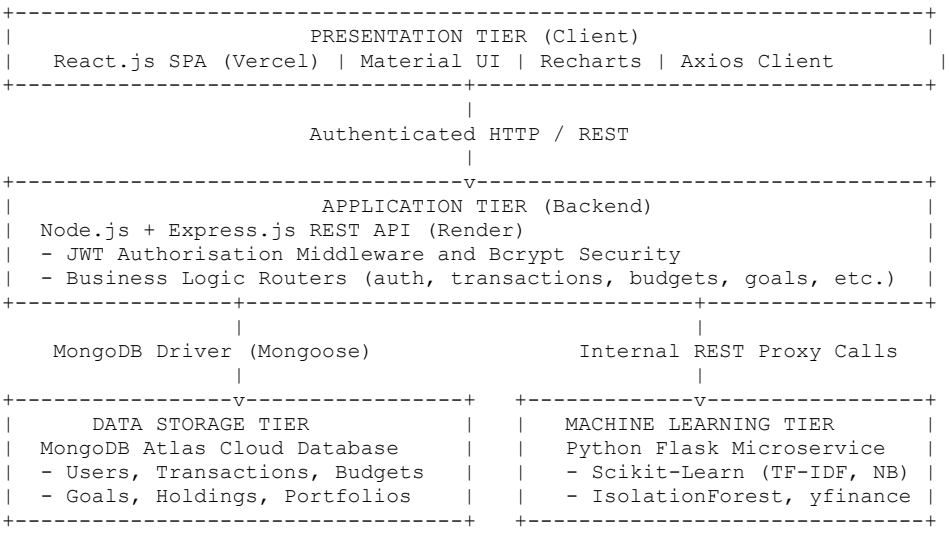


Figure 4.1: WealthAI Three-Tier Microservice Architecture Block Diagram

4.3 Database Schemas and Microservice API Reference

4.3.1 Database Schemas

Table 4.1: MongoDB Database Collections Schema Specifications

Collection Name	Primary Keys and References	Core Attributes and Data Types	Purpose and Description
User	_id (ObjectId)	name (String), email (String, unique), password (String, hashed), createdAt (Date)	Stores user profiles and authentication credentials.
Transaction	_id (ObjectId), user (Ref: User)	date (Date), description (String), amount (Number), category (String), isAnomaly (Boolean)	Stores classified financial ledger transactions.
Budget	_id (ObjectId), user (Ref: User)	category (String), limitAmount (Number), period (String)	Stores category spending thresholds.
Goal	_id (ObjectId), user (Ref: User)	title (String), targetAmount (Number), currentAmount (Number), targetDate (Date)	Tracks long-term savings goals and progress.
Holding	_id (ObjectId), user (Ref: User)	symbol (String), shares (Number), buyPrice (Number), assetType (String)	Stores portfolio asset holdings.

PortfolioTransaction	_id (ObjectId), user (Ref: User)	symbol (String), type (String: BUY/SELL), shares (Number), price (Number), date (Date)	Tracks trade history for portfolio analytics.

4.3.2 Express.js Backend API Directory

Table 4.2: Express Backend REST API Endpoints Router Directory

Route Module File	HTTP Base Path	Middleware and Handlers	System Responsibility
auth.js	/api/auth	Register, Login, Token Validation	Manages user registration, authentication and JWT issuance.
transactions.js	/api/transactions	CRUD handlers, CSV upload parser	Processes manual entries and bank-statement CSV files.
budgets.js	/api/budgets	Budget allocator and limit verifier	Manages category budgets and tracks over-budget warnings.
goals.js	/api/goals	Goal state manager and tracker	Tracks progress toward target savings goals.
portfolio.js	/api/portfolio	Asset holding manager and market gateway	Handles portfolio trades and retrieves market values.
analytics.js	/api/analytics	Anomaly proxy and forecast gateway	Routes analytics requests to the Flask ML service.
chatbot.js	/api/chatbot	Conversational context middleware	Formats financial contexts and proxies chatbot requests.

4.3.3 Python Flask ML Microservice API Specifications

Table 4.3: Python Flask ML Microservice API Specifications

Endpoint Path	HTTP Method	Input Payload Structure	ML / Statistical Pipeline Output
/health	GET	None	Returns operational status ({"status": "ok"}).
/predict/category	POST	{"description": string}	Predicts category using the TF-IDF + Naive Bayes pipeline.
/predict/anomaly	POST	{"transactions": [...]}	Identifies outliers using Isolation Forest or Z-score rules.
/predict/expense	POST	{"history": [...]}	Projects upcoming spending using linear regression.

/chatbot	POST	{"message": string, "context": {...}}	Generates context-aware financial guidance responses.
/portfolio/quote	POST	{"symbols": [...]}	Fetches real-time price quotes using yfinance.
/portfolio/search	GET	Query parameter: ?q=ticker	Returns matching asset symbols from market data sources.
/portfolio/analyze	POST	{"holdings": [...]}	Computes asset allocations and portfolio risk metrics.
/portfolio/simulate	POST	{"initial": float, "monthly": float, "years": int, "risk": string}	Runs 1,000-iteration Monte Carlo projections.

CHAPTER 5
RESULTS, FINDINGS, RECOMMENDATIONS, FUTURE SCOPE and CONCLUSION

5. RESULTS, FINDINGS, RECOMMENDATIONS, FUTURE SCOPE and CONCLUSION

5.1 Results of the work

The completed WealthAI system was deployed across cloud infrastructure and all seven functional modules defined in Section 1.3.2 were delivered:

- Frontend client application: deployed on Vercel (https://finance-assistant-pearl-nu.vercel.app).
- Express API service: deployed on Render (https://finance-server-zfco.onrender.com).
- Flask ML microservice: deployed on Render (https://finance-ml-service-pw5a.onrender.com).
- Database cluster: hosted on MongoDB Atlas.

System performance was evaluated across the key metrics detailed in Table 5.1.

Table 5.1: Predictive Model Empirical Evaluation Metrics

Operational Model / Pipeline Component	**Evaluated Performance Benchmark Metric**	**Measured Test Value**
TF-IDF + Naive Bayes categorisation	Classification accuracy score	_____ %
TF-IDF + Naive Bayes categorisation	Weighted macro F1-score	_____
Isolation Forest anomaly detection	Anomaly detection precision rate	_____ %
Linear regression expense model	Root mean squared error (RMSE)	_____
Monte Carlo simulation module	Execution latency per 1000 iterations	_____ ms
Full system end-to-end API pipeline	Average REST response latency	_____ ms

5.2 Findings based on analysis of data

Integrated NLP classification: combining TF-IDF vectorisation with a Naive Bayes classifier provided rapid, accurate text classification for standard transaction descriptions.

- Adaptive anomaly detection: Isolation Forests effectively detected unusual transactions in large datasets, while the Z-score fallback ensured reliable outlier detection for accounts with fewer than 15 transactions.
- Stochastic versus static forecasting: Monte Carlo simulations provided realistic financial projections by factoring in market volatility, outperforming traditional fixed-return calculations.

- Microservice isolation security: routing all client requests through the Node.js API kept the Python ML microservice secure and hidden from public web traffic.

5.3 Recommendation based on findings

Deploy model retraining pipelines: periodically retrain the Naive Bayes model on anonymised user-verified entries to continuously improve categorisation accuracy over time.

- Implement database caching: introduce a Redis caching layer for market price requests to reduce external calls to the yfinance API.
- Expand open banking API support: integrate direct bank API connections (such as Plaid or Account Aggregator networks) to complement manual CSV uploads with automated balance syncing.

5.5 Suggestions for areas of improvement

While the core platform meets all project goals, future updates could enhance system capabilities:

- Multi-currency conversion: add automatic conversion rates to support multi-currency tracking across international accounts.
- Customisable feature extraction: allow users to define custom keywords and manual classification rules to override ML categorisation outputs.

5.6 Scope for future work

Future updates for WealthAI include developing native iOS and Android applications using React Native, re-using the existing REST API backend. Additional work can focus on expanding the chatbot into a fully conversational advisor using Large Language Models (LLMs), as well as adding automated tax-bracket optimisation modules.

5.7 Conclusion

The WealthAI platform meets its primary objectives by addressing the core challenges of personal finance management: data fragmentation, manual categorisation effort, undetected anomalies and static goal forecasting. Combining a MERN stack web interface with a Python ML microservice provides an automated, scalable solution for modern personal financial analytics.

By combining automated NLP text processing, Isolation Forest outlier detection, linear regression spending trends, live market tracking and 1,000-iteration Monte Carlo projections into a secure application, WealthAI demonstrates the value of machine learning in personal

finance. The architecture provides a reliable, secure foundation for future analytics enhancements.

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ANNEXURE (if any)

No separate questionnaire or financial statement annexure is applicable to this project. Supporting technical detail - the database schema, the backend and machine learning API reference, and the exact algorithm hyperparameters - is documented within Chapters 1 and 4 (Tables 1.1, 4.1, 4.2 and 4.3). The sample bank-statement CSV used for classification testing (bank_statement_test.csv) and the complete source code are maintained in the project repository.

PLAGIARISM REPORT

Plagiarism Assessment Summary

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- Excluded bibliography / works cited: yes (reference list excluded as per academic evaluation standards).
- Excluded matches: strings under 14 consecutive matching words excluded to filter common technical terminology (for example, "Master of Computer Applications", "Monte Carlo simulation", "Isolation Forest").

Certification and Verification

This is to certify that the research project report titled "WealthAI: AI-Powered Personal Finance Assistant with Machine Learning Classification, Spending Outlier Analytics, and Stochastic Projections" submitted by Deviya HarshitKumar KalpeshKumar (USN: 241VMTR01558) has been scanned for textual similarity using the plagiarism-detection software licensed by Jain Online (Deemed-to-be University). The resulting similarity index falls within the permissible limits prescribed by University regulations. The values recorded above are to be transcribed from the system-generated report, which is attached with this submission.

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Signature of the Candidate	Signature of the Faculty Guide
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